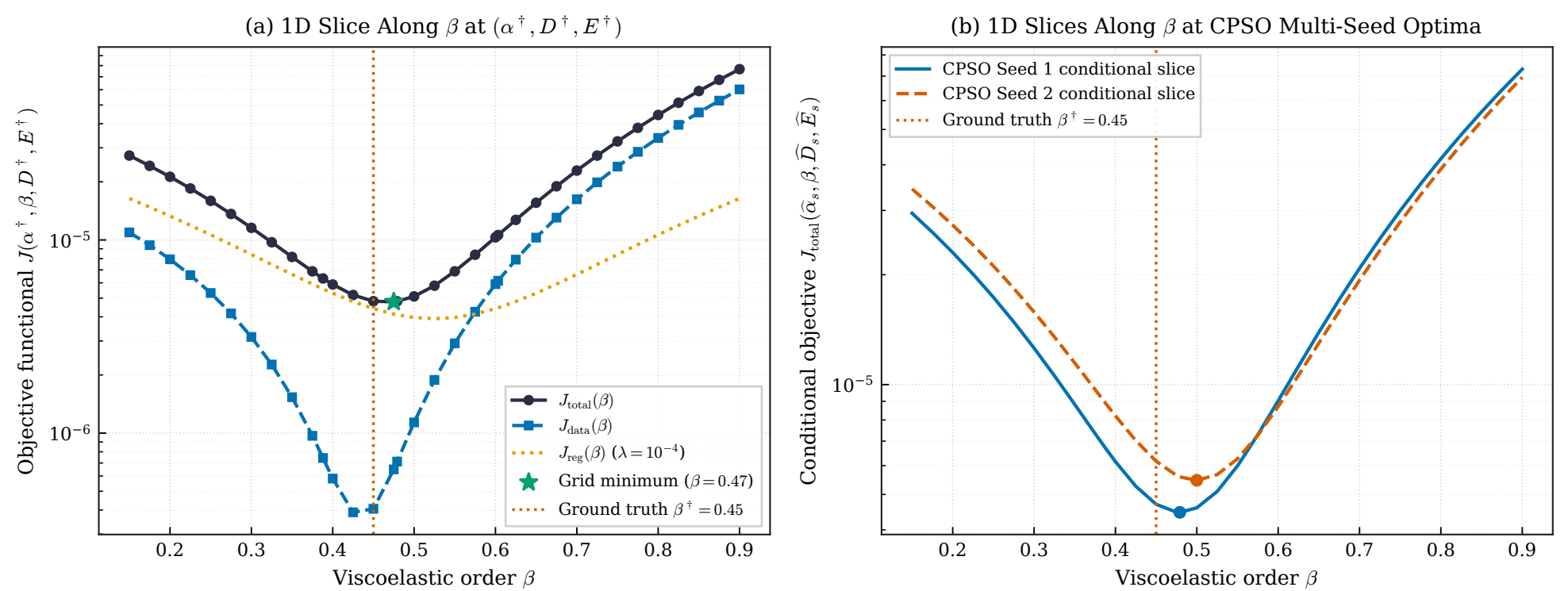
(a) 1D Slice Along β at $(\alpha^\dagger, D^\dagger, E^\dagger)$
Objective functional $J(\alpha^\dagger, \beta, D^\dagger, E^\dagger)$
10^{-5}
10^{-6}
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Viscoelastic order β
$J_{\text{total}}(\beta)$
$J_{\text{data}}(\beta)$
$J_{\text{reg}}(\beta)$ ($\lambda = 10^{-4}$)
Grid minimum ($\beta = 0.47$)
Ground truth $\beta^\dagger = 0.45$
(b) 1D Slices Along β at CPSO Multi-Seed Optima
Conditional objective $J_{\text{total}}(\widehat{\alpha}_s, \beta, \widehat{D}_s, \widehat{E}_s)$
10^{-5}
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Viscoelastic order β
CPSO Seed 1 conditional slice
CPSO Seed 2 conditional slice
Ground truth $\beta^\dagger = 0.45$